

## Unit 11, Lesson 6: Interpreting Linear Equations Fit to Bivariate Data

### Benchmarks

MA.8.AR.3.5 Given a real-world context, determine and interpret the slope and  $y$ -intercept of a two-variable linear equation from a written description, a table, a graph, or an equation in slope-intercept form.

MA.8.DP.1.2 Given a scatter plot within a real-world context, describe patterns of association.

### Learning Targets

- ☐ I can describe patterns of associations in data on a scatter plot.
- ☐ I can determine the slope and  $y$ -intercept from a given line fit to a set of bivariate data.
- ☐ I can interpret the slope and  $y$ -intercept of a linear equation in a real-world context.

11.6.1 Warm-Up: Number Puzzles

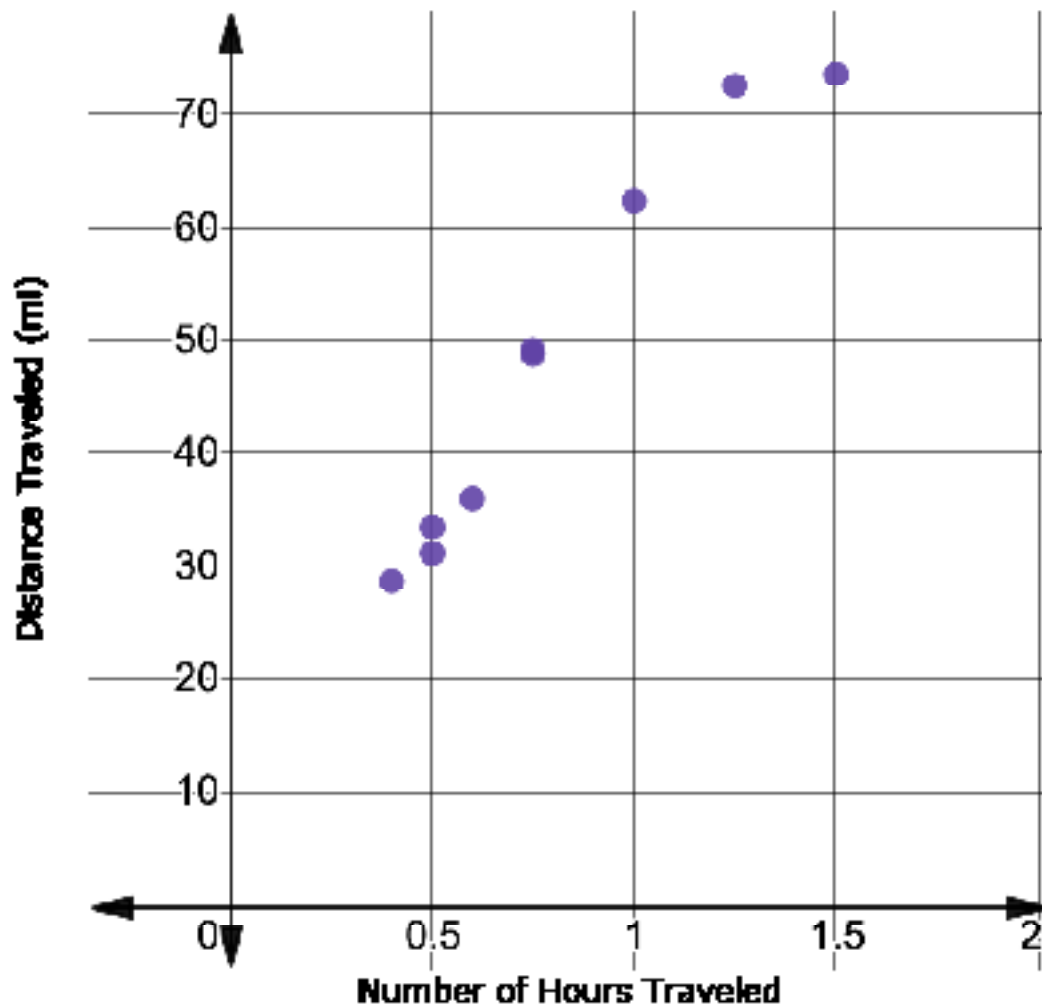
1. Read the clues for each puzzle to find the answer.

|          | Clues                                                                                                                                                             | Answer |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Puzzle 1 | <ul style="list-style-type: none"><li>• My value is even.</li><li>• My value is greater than 6.</li><li>• My value is less than 10.</li></ul>                     |        |
| Puzzle 2 | <ul style="list-style-type: none"><li>• My value is even.</li><li>• My value is not less than 5.</li><li>• My value is less than 8.</li></ul>                     |        |
| Puzzle 3 | <ul style="list-style-type: none"><li>• I am a multiple of 10.</li><li>• I am greater than <math>4^2</math>.</li><li>• I am less than <math>5^2</math>.</li></ul> |        |

2. What strategies did you use to determine the answer for each puzzle?

**11.6.2 Exploration: Linear Equations Modeling Bivariate Data**

1. Camilo traveled with his family from Jasper, FL to Hialeah, FL. The trip was 436 miles using I-95 to avoid tolls. Throughout the trip, Camilo recorded the distance traveled, in miles (mi), for specific amounts of time, in hours. The scatter plot shows the data Camilo recorded.



- a. The point  $(0.5, 33.5)$  is on the scatter plot. Complete the statement to interpret the meaning of this point in context.

For one of the times Camilo recorded their travel, the family had

traveled

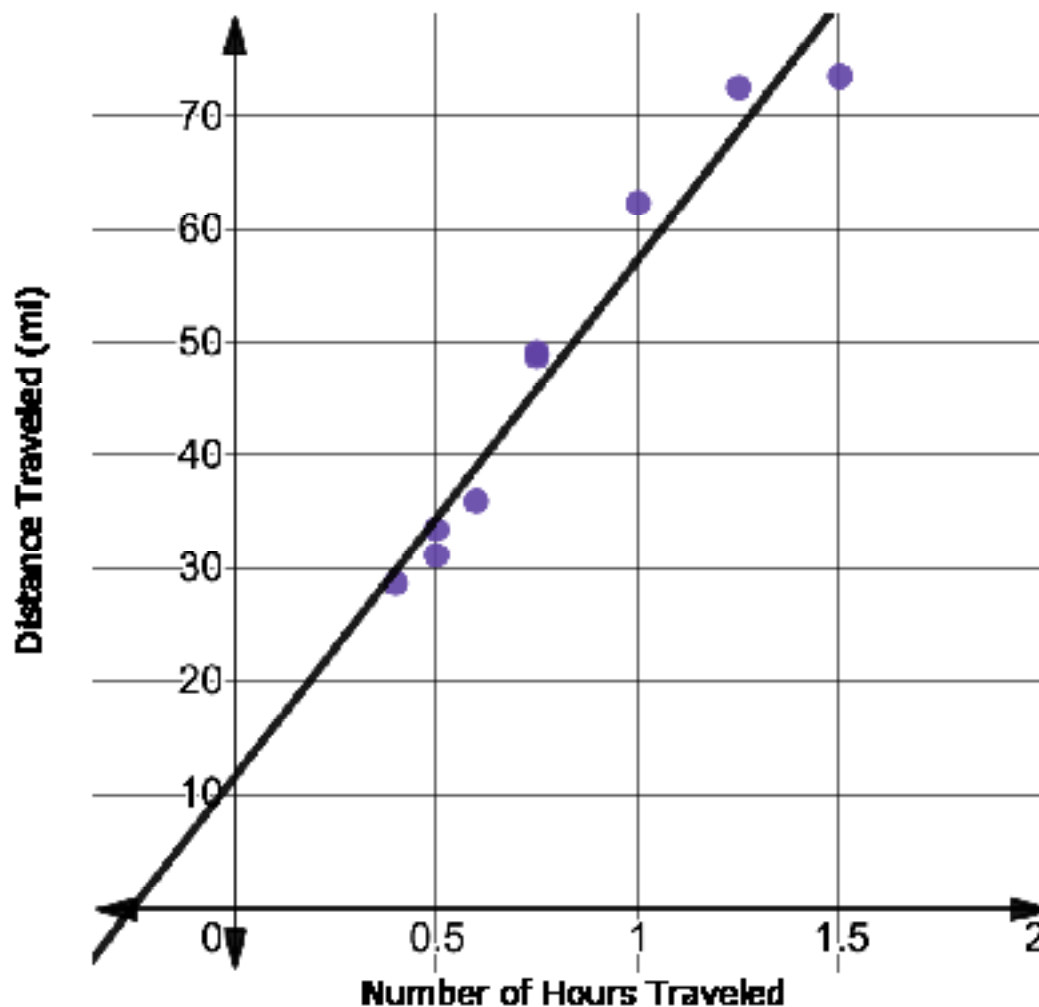
- ☐ 33.5 miles  
☐ 33.5

in

- ☐ 0.5 mile.  
☐ 0.5 hour.

- b. At what speed was the family traveling during the period of time represented by the point in Part A?
- c. Describe the association between the number of hours traveled and distance traveled by the family.

2. Suppose the line shown is fit to the data.



The equation of the line of fit is  $y = 45.9x + 11.5$ .

- a. Discuss with your partner what you think this line of fit represents in this context. Summarize your discussion.

- b. Complete the statement.

The slope of the line fit to the data is

- ☐ 11.5.
- ☐ 45.9.

This means, on average, the family traveled about

- ☐ 11.5
- ☐ 45.9

- ☐ miles per hour
- ☐ hours per mile

on their trip from Jasper to Hialeah.

- c. Discuss with your partner whether you think the  $y$ -intercept of the line makes sense for this context.

### **11.6.3 Bring It Together: Linear Equations Modeling Bivariate Data**

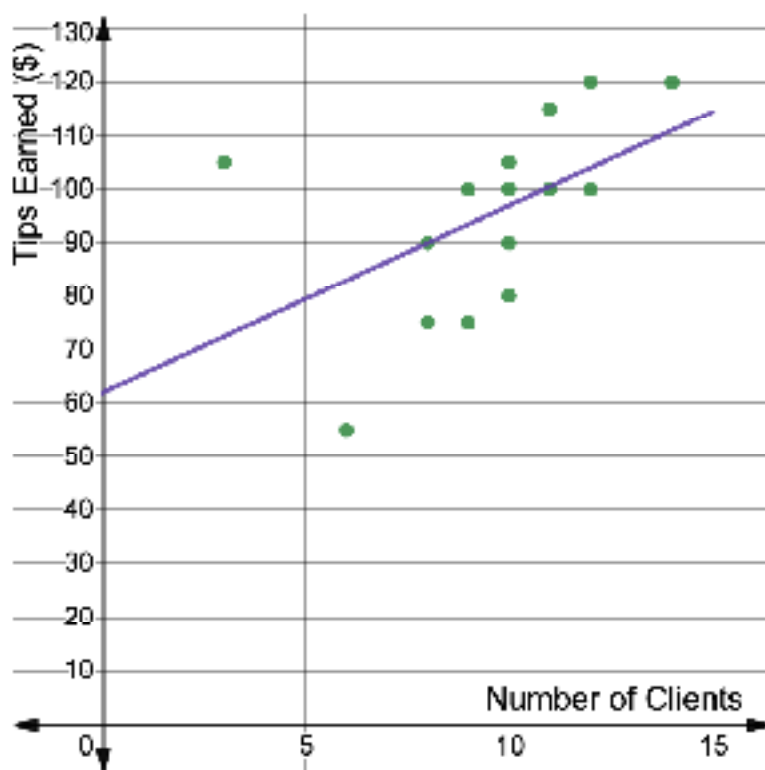
**When a linear equation is fit to bivariate data in a scatter plot, the line approximates the relationship between two sets of data to represent the overall trend.**

**The slope and y-intercept of the line can be interpreted in context, but there are times when the interpretation may not make sense in context because it is representing an estimate based on a set of data.**

**11.6.4 Guided Practice: Interpreting Linear Equations Fit to Bivariate Data**

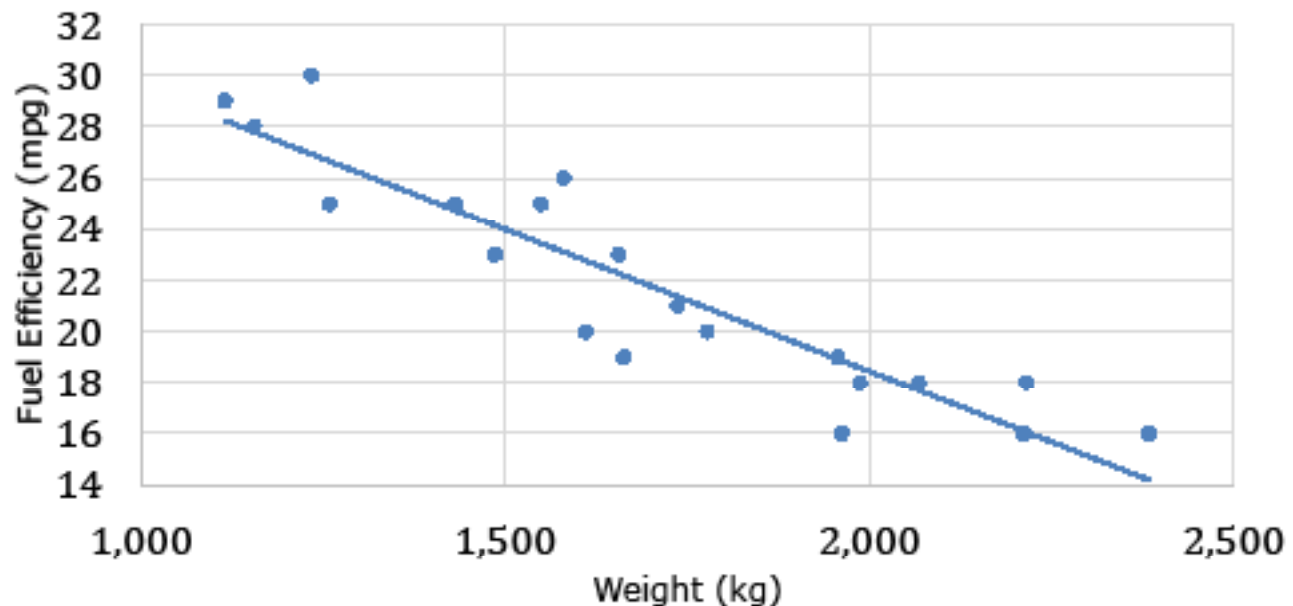
1. Aditya works as a hair stylist and was looking at his earnings for the past few weeks. He created the scatter plot shown to see if there was a relationship between how many clients he saw in a day and how much he earned in tips.

The equation of the line of fit is  $y = 3.5x + 61.94$ .



- What is the slope of the line fit to the data?
- Interpret the meaning of the slope in context.
- Explain why the  $y$ -intercept of the line estimating the relationship between the variables does not make sense in this context.

2. The scatter plot shown represents the relationship between a car's weight, in kilograms (kg), and its fuel efficiency, in miles per gallon (mpg).



The equation of the line of fit is  $y = -0.012x + 41.3$ .

- a. Complete the statement.

The ☐ weight  
☐ fuel efficiency of the car ☐ increases  
☐ decreases by about ☐ 0.012  
☐ 41.3

☐ kg  
☐ mpg as the ☐ weight  
☐ fuel efficiency of the car ☐ increases  
☐ decreases

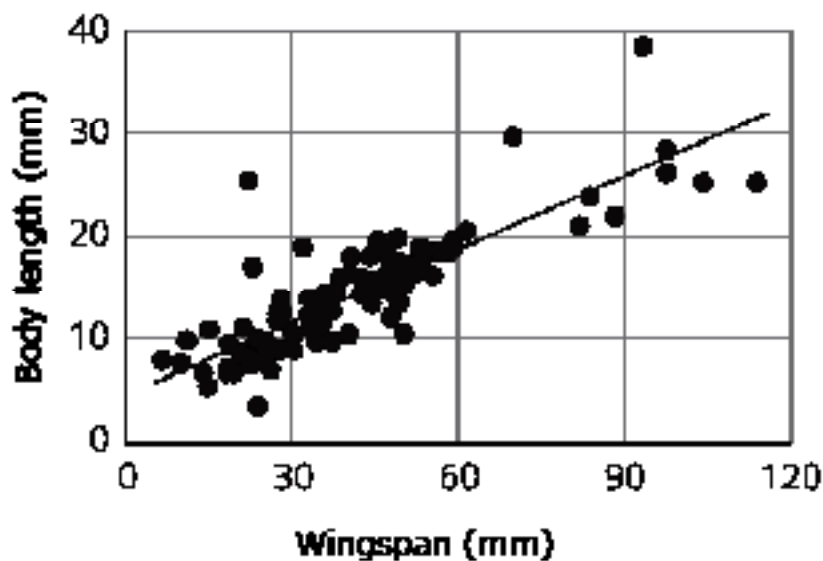
by 1 ☐ kg.  
☐ mpg.

- b. Describe how the direction of the association of the data relates to the slope of the line fit to the data.



**11.6.5 Your Turn!**

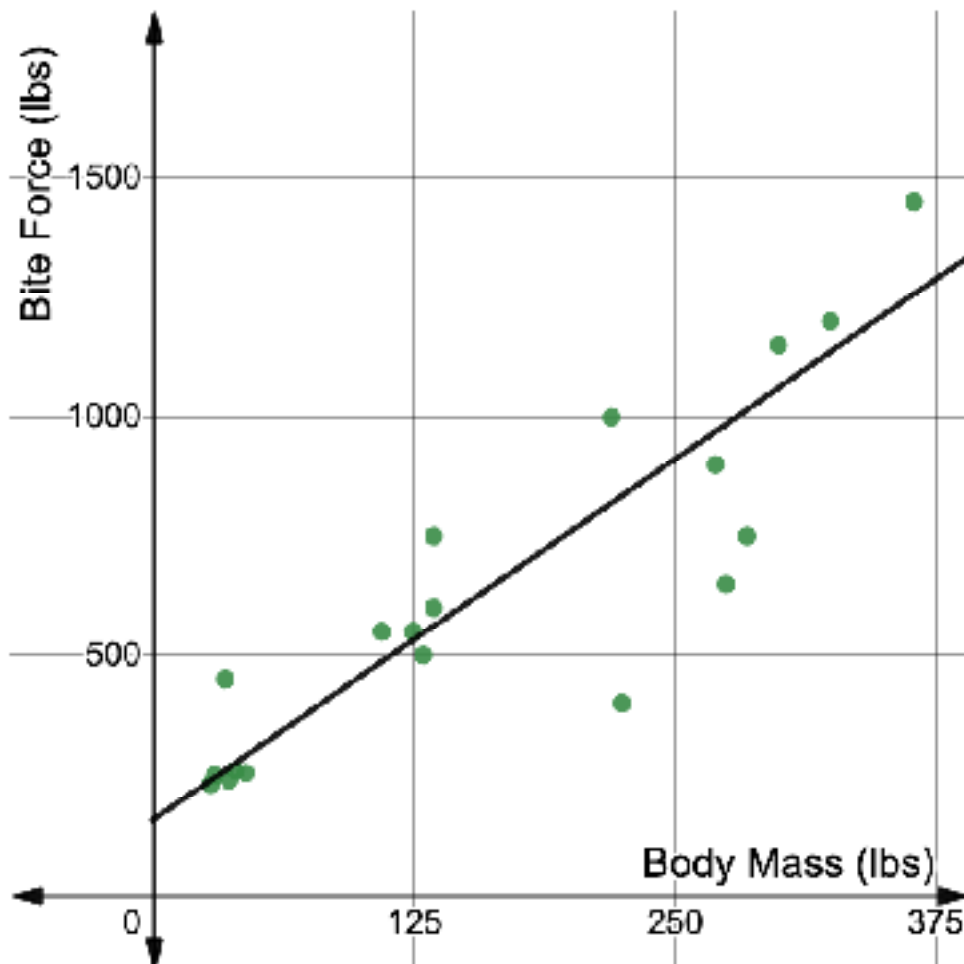
1. Researchers have collected data comparing the wingspan and body length of butterflies, in millimeters (mm). The scatter plot represents the data collected with the line,  $y = 0.238x + 4.642$  fit to the data.



Interpret the slope of the equation modeling the relationship.

**11.6.6 Wrap-Up: Ticket Out the Door**

1. Zoologists studied different species of animals to examine if there was any relationship between the body mass of the animal, in pounds (lbs), and the animal's bite force, in lbs.



The equation of the line of fit is  $y = 3.02x + 154.7$ .

- a. What is the slope of the line fit to the data?
- b. Interpret the meaning of the slope in context.